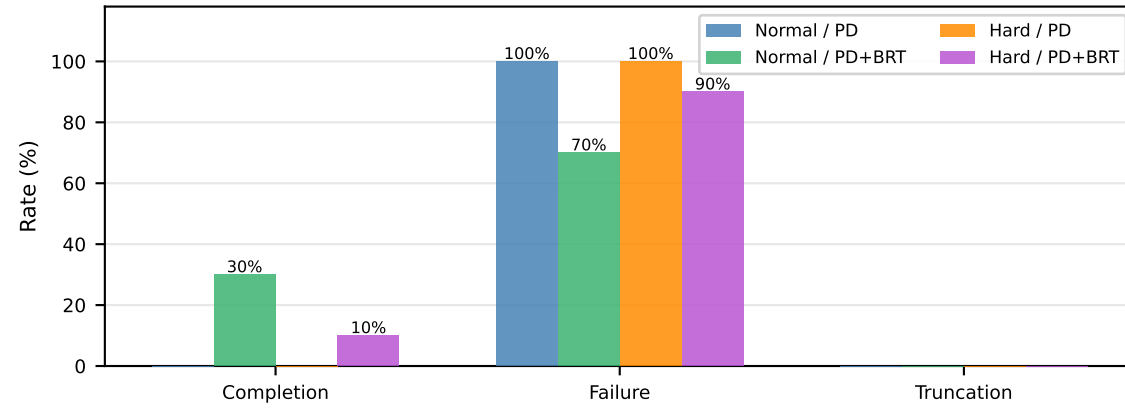
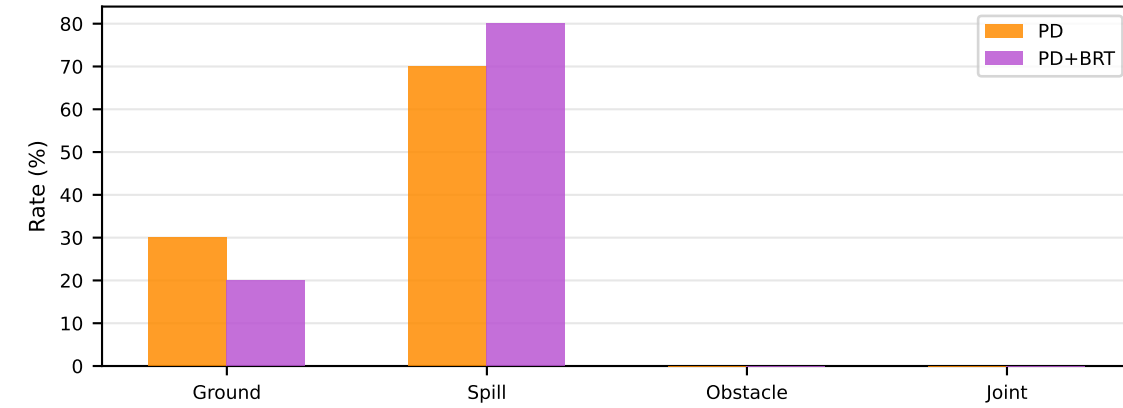


Safety evaluation: PD controller vs PD + BRT safety filter (Kp=5.0, Kd=3.0)
PD hard-start failure 100% | PD+BRT hard-start failure 90%

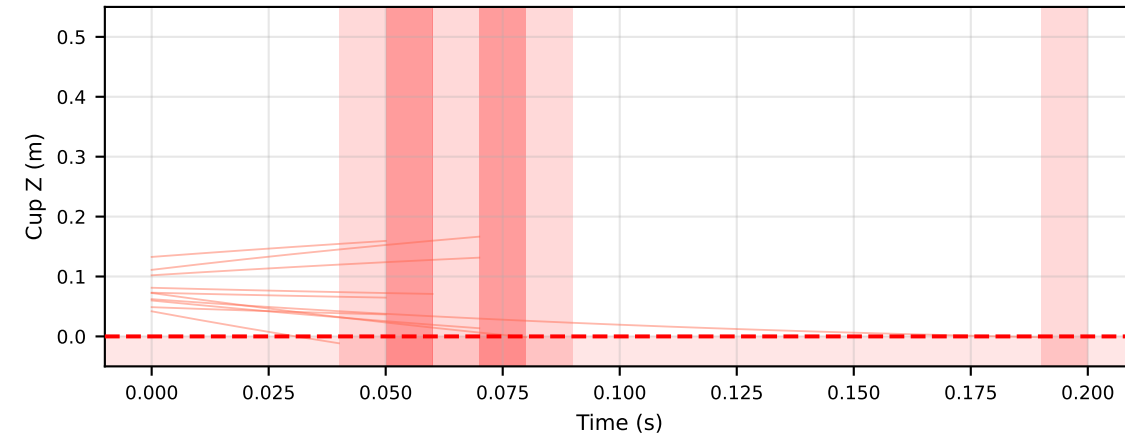
(A) Episode outcomes



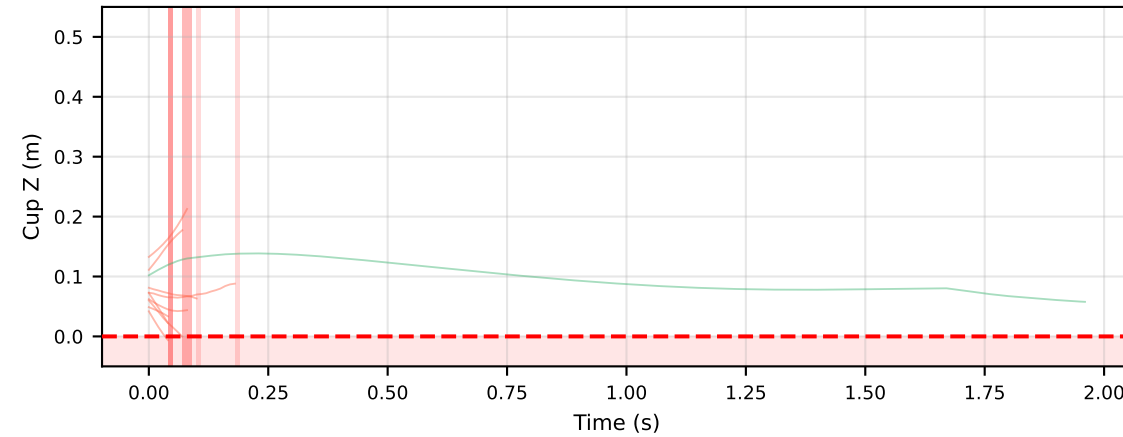
(B) Failure type breakdown (hard starts)



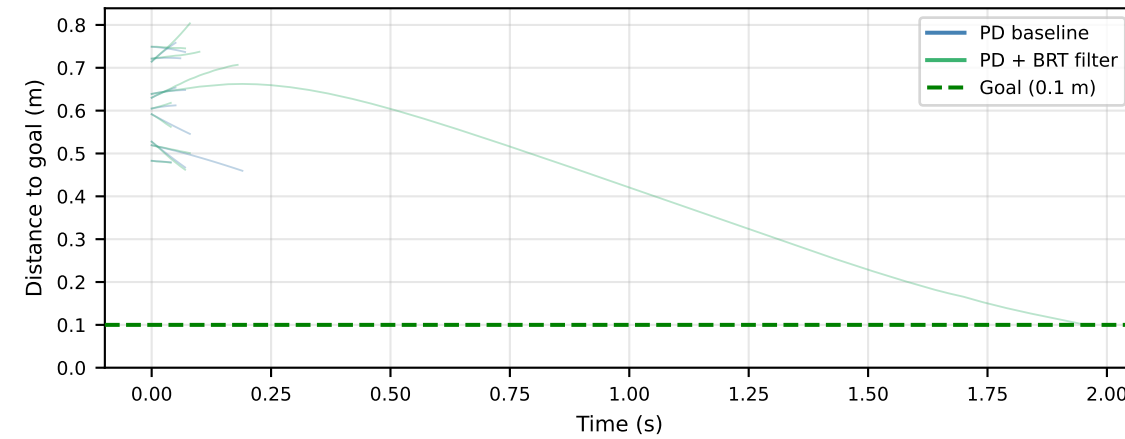
(C) Hard starts — PD baseline (cup height)



(D) Hard starts — PD + BRT filter (cup height)



(E) Progress toward goal (hard starts)



(F) BRT filter activity (hard starts)

